

SATYA AKHIL GALLA

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EDUCATION

Boston University

Master of Science in Artificial Intelligence

Jan 2026

GPA: 3.7/4.0

IIIT Sri City

Bachelor of Technology in Computer Science

May 2024

GPA: 3.5/4.0

TECHNICAL SKILLS

Generative AI: Agents (LangGraph), Neuro-Symbolic AI, Llama-3, RAG, Transformers, Phi-3 (Quantization), Vector DBs

Engineering: Docker, FastAPI, AWS (EC2/S3), TCP/IPC Sockets, Git, CI/CD, SQL, Bash Scripting, Linux

Data Science: XGBoost, Time-Series Forecasting, GCNs, Regex Parsing, Pandas, NumPy, Scikit-learn, OpenCV

EXPERIENCE

ISRO (Indian Space Research Organisation)

Machine Learning Engineer Intern

Dec 2022 – May 2023

India

- Authored **IEEE WHISPERS 2023** paper presenting the first hyperspectral classification framework for Chandrayaan-2 IIRS data, utilizing a **Latent GCN** approach.
- Engineered a **Spectral-Spatial Non-Linearity** graph formulation to handle irregular lunar topography, achieving **90.1% accuracy** on Chandrayaan-1 (M3) benchmarks.

Aaizel International Tech

Machine Learning Engineer Intern

Feb 2024 – May 2024

India

- Created the domain's **first remote sensing scene graph dataset** and architected a multimodal fusion pipeline (IFCNN) merging IR+RGB satellite imagery.
- Fine-tuned a **RelTR Transformer** for geospatial relationship extraction and deployed a **Gradio** interface for instant semantic visualization.

Terrafic Inc

Computer Vision Intern

June 2023 – Nov 2023

India

- Implemented Super-Resolution (SR) algorithms (4x/6x) to enhance object detection on low-resolution defense mapping data.
- Integrated **Segment Anything Model (SAM)** to automate maritime feature extraction, delivering a production-ready MVP that secured initial client pilots.

Boston University

Graduate Teaching Assistant – CS 365 Foundations of Data Science

Sep 2025 – Present

USA

- Lead technical labs on Python optimization and statistical modeling, mentoring 50+ students in data science fundamentals.
- Facilitate weekly code reviews and debugging sessions to reinforce best practices in algorithmic implementation.

KEY PROJECTS

Retrieval-Augmented Generation (RAG) for Healthcare | Llama-3, LangChain | GitHub

Sep 2024 – Dec 2024

- Architected a RAG system indexing **207,000+** verified medical Q&A pairs, engineering a batch ingestion pipeline to process data in **<2 hours**.
- Achieved an **87% SBERT alignment score** to reduce hallucinations, ensuring high semantic fidelity against verified references.

Cozy Companion (CoCo): Neuro-Symbolic AI Agent | Phi-3, Godot | GitHub

Jan 2025 – Present

- Engineered a **Subsumption Architecture** within **Godot**, decoupling a 60Hz reactive loop from asynchronous **Phi-3 (4-bit)** LLM inference via TCP sockets to achieve **<16ms latency**.
- Integrated a **Leaky Integrate-and-Fire (LIF)** model to estimate user flow state from interaction patterns, dynamically modulating agent behavior.

Visual Question Answering (LRCN) | PyTorch, VQA-v2 | GitHub

Jan 2025 – May 2025

- Reproduced LRCN architecture by translating mathematical paper formulations directly into optimized PyTorch modules.
- Benchmarked performance against SOTA, achieving results within **0.1% parity** of the original publication.

Predictive Analytics for MBTA Transit | XGBoost, GIS | GitHub

Feb 2024 – May 2024

- Engineered a unified ETL pipeline aggregating **9 years** (2015-2024) of multi-modal data to train **XGBoost** (Reliability) and **Random Forest** (Ridership) models.
- Uncovered a system-wide **33.35% post-pandemic ridership drop** and identified critical service disparities in 30-60% AMI (Area Median Income) corridors.

Multimodal Skin Cancer Detection System | PyTorch | GitHub

Sep 2024 – Dec 2024

- Tackled 0.1% class imbalance using synthetic oversampling and developed a voting ensemble fusing CNNs and tabular models.
- Achieved an **ROC-AUC of 0.96**, surpassing single-modal baselines by **15 points** in final validation.

Generative Audio: Transformer vs. Diffusion | AudioLDM | GitHub

Jan 2025 – May 2025

- Conducted a comparative evaluation of **8 hybrid pipelines**, identifying a trade-off between semantic alignment (Transformers) and audio fidelity (Diffusion).
- Optimized a cascaded architecture to achieve a peak **CLAP score of 0.35** while maintaining high fidelity with a **Frechet Audio Distance (FAD) of 6.55**.

SELECT PUBLICATIONS

Efficient Graph Formulation and Latent Space Integration for Lunar Hyperspectral Image Classification – IEEE WHISPERS 2023

Adaptive Graph Construction for GCNs – NASA NESF 2023 (Poster)